Fig. 5 | Spatial velocity, scalability and read-depth stability. a, Spatial velocity on the MPSTA ST_small subset (608 spots), with 19 available method fields. Cell-marker radius is increased 1.6-fold for visibility. TopicVelo uses five native stochastic topic transitions rerun from saved fits after U/S signal QC, replacing the historical last-topic-only configuration. Its GDC (0.484) uses T @ Ms - Ms as a transition- displacement proxy; the spatial field is projected from native transitions. This is not its historical base-scVelo output. The first slot shows the five directed transitions used as benchmark references: Inner EPC 1 → Inner EPC 2 → Outer EPC 1 → Outer EPC 2, and Inner EPC 2 → Secondary P-TGC1 → Secondary P-TGC2. The labeled reference nodes also provide the cell-type color key. Cell-type nomenclature and reference transitions follow the original MPSTA study (Wu et al., Cell Discovery, 2024). Positions in this reference diagram are schematic, not spatial coordinates. TFvelo has distinct preprocessing; TopicVelo uses the documented rerun above; the other method panels are historical. **b,** Runtime across 1,000–60,000 cells, with labels rounded up to minutes or hours and colors capped at 60 minutes. Blank cells indicate unavailable measurements. **c,** Mean per-cell velocity cosine relative to full-depth predictions (1.0× reads). The 1.0× point is a self-comparison; reduced-depth points assess prediction consistency, not biological accuracy. The added cell2fate results use 9,144 shared cells and 472 shared genes. TopicVelo denotes integrated-transition displacement in fixed full-depth common-gene Ms coordinates, not its base scVelo velocity; its historical single-topic stochastic configuration is retained. Display order: **a,** descending GDC down each column, then from left to right, excluding the reference schematic; **b,** ascending mean runtime across the 11 shared tested cell counts from 1K to 20K from left to right; **c,** legend ordered by descending cosine at reads ratio 0.2, from left to right and then top to bottom.